

Indian Stock Market Microstructure Analysis

Research Report

Internship Capstone Project
Data Science / Machine Learning Track
[Your Name] | [Your Institution] | [Date]

Abstract

This report presents a research study of Indian equity market behavior using publicly available daily price and volume data for the NIFTY 50 index (or a selected NSE-listed stock/index). Because true tick-level order-book data is impractical to obtain for a beginner-scale project, this study uses daily Open-High-Low-Close-Volume (OHLCV) data as a simplified proxy for market microstructure. The analysis combines exploratory data analysis, statistical hypothesis testing, and unsupervised machine learning (clustering) to characterize how return, volatility, and trading volume interact, and to identify distinct, data-driven “market regimes” within the historical price series. A supervised classifier is additionally used to validate the internal consistency of the discovered regimes. All findings in this report are descriptive and historical in nature; the study does not propose or evaluate any trading strategy.

1. Introduction

1.1 Background

Market microstructure is the branch of financial economics concerned with the process by which investors' latent demands are translated into transactions — prices, quotes, order flow, and trading volume. It seeks to explain why, even absent new fundamental information, prices and trading activity fluctuate at high frequency, and how liquidity and volatility are related. Traditionally, microstructure research relies on tick-by-tick trade and quote data, which is costly and complex to obtain and process.

At the same time, several of microstructure's central empirical themes are visible even in daily data — most notably the well-documented positive relationship between trading volume and price volatility, and the tendency for volatility to cluster in time (periods of high volatility are followed by more high volatility, and calm periods persist). This project treats daily OHLCV data as a tractable, beginner-accessible proxy for studying these themes in the context of the Indian equity market.

1.2 Problem Statement

Retail-accessible research on Indian market behavior rarely goes beyond simple technical-analysis narratives. There is comparatively little beginner-oriented, statistically rigorous, and reproducible analysis that (a) formally tests whether well-known microstructure relationships (e.g., volume–volatility) hold for Indian indices/stocks, and (b) attempts to algorithmically discover distinct behavioral regimes in the data, rather than assuming regimes based on calendar periods or news narratives.

1.3 Objectives

Characterize the statistical properties of returns, volatility, and volume for the selected Indian index/stock.

Formally test the hypothesis that trading volume is associated with the magnitude of price movements.

Engineer a feature set capable of describing each trading day's "behavior" (not just its price level).

Apply unsupervised clustering (KMeans and Hierarchical Clustering) to discover natural market regimes from these features.

Validate the discovered regimes using a simple supervised classifier and standard evaluation metrics.

Report findings, limitations, and future research directions in a transparent, reproducible manner.

1.4 Scope and Non-Goals

This study is explicitly a descriptive/behavioral research project. It does not: (a) forecast future prices or returns, (b) propose or backtest a trading strategy, or (c) make any investment recommendation. Regime labels discovered via clustering describe historical structure only and carry no guarantee of persistence into the future.

2. Conceptual and Literature Background

2.1 Volume–Volatility Relationship

A substantial body of finance literature documents a positive relationship between trading volume and the magnitude of price changes. Early theoretical and empirical work (e.g., Clark, 1973; Karpoff, 1987) proposed the "mixture of distributions hypothesis", under which both returns and volume are jointly driven by the same underlying rate of information arrival – on days with more new information, both price movement and trading activity tend to be larger. This motivates Hypothesis Test 1 in Section 4.

2.2 Volatility Clustering and Fat Tails

Mandelbrot (1963) and later Engle's ARCH/GARCH family of models formalized the empirical observation that "large changes tend to be followed by large changes, of either sign, and small changes tend to be followed by small changes" – i.e., volatility clustering. Relatedly, asset returns typically exhibit heavier tails than a Normal

distribution (extreme days occur more often than a Gaussian model would predict). Both phenomena are examined descriptively in this study's exploratory data analysis (Section 5).

2.3 Market Regimes and Regime-Switching Models

The idea that markets alternate between qualitatively distinct states — e.g., calm/trending versus turbulent/mean-reverting — underlies regime-switching models such as Hamilton's (1989) Markov-switching model and, in machine learning contexts, clustering-based regime identification. This project uses KMeans and Hierarchical Clustering as an accessible, beginner-appropriate approximation of formal regime-switching modeling, discussed further as a future extension (Section 8, and see the companion Future Scope notes).

2.4 Positioning of This Study

This project does not aim to contribute novel financial theory. Its contribution is applied and pedagogical: demonstrating, on real Indian market data, how established microstructure themes can be investigated end-to-end using an accessible, reproducible data science workflow (cleaning → EDA → feature engineering → hypothesis testing → clustering → validation).

3. Data and Methodology

3.1 Data Source

Historical daily OHLCV data is sourced from Yahoo Finance via the yfinance Python library. The primary instrument analyzed is the NIFTY 50 index (ticker “^NSEI”), with BANK NIFTY (“^NSEBANK”) and individual NSE-listed stocks (e.g., “RELIANCE.NS”) supported as direct substitutions.

Field	Description
Ticker	^NSEI (NIFTY 50) — configurable
Period	[START_DATE] to [END_DATE] — fill in after running the notebook
Frequency	Daily
Fields	Open, High, Low, Close, Adj Close, Volume

3.2 Data Preparation

Raw data is inspected for shape, data types, missing values, and duplicate rows.

Missing values are handled via forward-fill (carrying the last known value forward), which respects the temporal ordering of the data and avoids introducing lookahead bias. Rows violating basic price logic (e.g., High lower than Low) are treated as data errors and removed. All cleaning decisions are documented in the companion analysis notebook.

3.3 Feature Engineering

The following behavioral features are engineered from the cleaned OHLCV data for use in clustering and hypothesis testing:

Daily Return and Log Return – percentage/logarithmic day-over-day price change.

Price Range and Price Range (%) – intraday High–Low spread, in absolute and normalized terms.

Rolling Mean and Rolling Std (20-day) – short-term trend and price dispersion.

Rolling Volatility (21-day) – standard deviation of daily returns over approximately one trading month.

Moving Averages (20-day, 50-day) – trend-following smoothing of price.

Lag Features – prior 1–3 days' returns and prior day's volume, to capture short-term memory.

RSI (14-day) – Relative Strength Index, a bounded momentum indicator.

3.4 Statistical Hypothesis Testing

Three complementary tests examine the volume–volatility relationship described in Section 2.1:

Independent two-sample T-test (Welch's) comparing mean absolute return between high- and low-volume days (split at the median).

One-way ANOVA comparing mean absolute return across three volume tertile groups (Low / Medium / High).

Pearson correlation (with significance test) between volume and absolute return.

3.5 Clustering Methodology

Engineered features are standardized (zero mean, unit variance) prior to clustering, since KMeans and Hierarchical Clustering are distance-based and otherwise would be dominated by large-scale features. The number of clusters (K) is chosen using the Elbow Method (inertia vs. K) together with the Silhouette Score, evaluated over K = 2 to 10. KMeans (with multiple random initializations) and Agglomerative Hierarchical

Clustering (Ward linkage) are both fit at the chosen K, and their assignments are cross-tabulated to assess agreement between methods.

3.6 Supervised Validation and Evaluation

A Logistic Regression model and a depth-limited Decision Tree are trained to predict a day's cluster label from that same day's features, using a chronological (non-shuffled) train/test split to respect the time-ordered nature of the data. This step validates that discovered regimes have learnable, consistent boundaries in feature space; it is explicitly not a forecasting exercise. Evaluation uses accuracy, per-class precision/recall/F1 (via `classification_report`), and a confusion matrix.

4. Hypotheses

H1 (T-test)

H_0 : Mean absolute daily return is equal for high-volume and low-volume days. H_1 : Mean absolute daily return differs between the two groups.

H2 (ANOVA)

H_0 : Mean absolute daily return is equal across Low/Medium/High volume tertiles. H_1 : At least one tertile's mean differs from the others.

H3 (Correlation Significance)

H_0 : The true Pearson correlation between volume and absolute return is zero. H_1 : The true correlation is non-zero.

Significance is assessed at the conventional $\alpha = 0.05$ threshold throughout.

5. Results (Template)

Results below are presented as a template to be completed with actual output once the accompanying notebook is executed on live data (this report was authored without live internet access to Yahoo Finance).

Test	Statistic	p-value	Conclusion
T-test (volume groups)	[t = ...]	[p = ...]	[Reject / Fail to reject H_0]
ANOVA (volume tertiles)	[F = ...]	[p = ...]	[Reject / Fail to reject H_0]
Pearson correlation	[r = ...]	[p = ...]	[Reject / Fail to reject H_0]

Clustering: optimal $K = [K]$, silhouette score = [value]. KMeans/Hierarchical agreement: [strong/moderate/weak] (see crosstab in notebook). Regime profiles: [summarize each cluster's characteristic volatility/return/RSI levels].

Supervised validation: Decision Tree test accuracy = [X]%, macro-F1 = [value]. Top features by importance: [list].

6. Discussion

[To be completed with live results.] Discuss whether the volume–volatility relationship found matches the literature reviewed in Section 2.1; whether volatility clustering/fat tails were visually and statistically evident; how many distinct regimes emerged and whether their characteristics align with intuitive market conditions (e.g., a clearly “calm” regime and a clearly “turbulent” regime); and whether the supervised classifier's high validation accuracy supports treating the regimes as internally consistent.

7. Limitations

Daily OHLCV data is a simplified proxy for true tick/order-book microstructure; findings should not be generalized to intraday microstructure phenomena (e.g., bid-ask spread dynamics).

Clustering describes historical structure only; regimes are not guaranteed to persist or recur in the same form.

The supervised learning step validates same-day feature-to-regime consistency, not forecasting ability.

Findings are specific to the analyzed ticker and date range and may not generalize across instruments or periods without re-running the analysis.

Standard statistical tests (T-test, ANOVA, Pearson correlation) assume certain distributional properties (e.g., independence, approximate normality of test statistics under the CLT for large samples) that were not exhaustively verified beyond the fat-tail check in EDA.

8. Conclusion

This study demonstrates a complete, reproducible workflow for investigating market microstructure themes in Indian equity data using only publicly available daily data and standard open-source tooling. Once executed on live data, it is expected to provide empirical, Indian-market-specific evidence on the volume–volatility relationship and to

yield a small number of interpretable, statistically distinguishable market regimes. The accompanying Jupyter notebook, final report, and insights document provide the full technical implementation, complete results, and a synthesis of practical takeaways, respectively.

References

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